

EXECUTIVE BRIEF

TO: Executive Leadership & Board of Directors

FROM: Cyber Risk & Compliance Team

DATE: July 30, 2026

SUBJECT: Business Impact Analysis (BIA) Key Findings & Risk Mitigation Strategy

Bottom Line Up Front (BLUF):

A failure in ApexPay’s core **API Payment Processing** service exposes the company to an estimated **\$45,000 per hour in direct revenue loss** and immediate contractual SLA breaches. Current architectural single points of failure threaten our target 15-minute recovery objective during a regional cloud outage. An immediate infrastructure enhancement is recommended to protect enterprise revenue and brand trust.

1. Risk Exposure Summary

Our recent Business Impact Analysis (BIA) categorized ApexPay's core operations into four critical tiers based on maximum tolerable downtime:

Business Function	Criticality Tier	Target RTO	Financial Impact / Hr	Key Risk Driver
API Payment	Tier 1	15	\$45,000	Direct revenue

Business Function	Criticality Tier	Target RTO	Financial Impact / Hr	Key Risk Driver
Processing	(Mission-Critical)	Minutes		loss, merchant churn.
User Authentication	Tier 1 (Mission-Critical)	1 Hour	\$25,000	Complete merchant portal lockout.
Fraud Detection Engine	Tier 2 (Business-Critical)	4 Hours	\$15,000	Increased exposure to fraudulent activity.
Merchant Settlement	Tier 2 (Business-Critical)	8 Hours	\$5,000	Regulatory non-compliance & penalties.

2. Key Findings & Strategic Gaps

1. Single Point of Failure (SPOF) in Primary Database:

The primary PostgreSQL database supporting API Payment Processing is confined to a single AWS region without automated cross-region failover. In a regional cloud outage, recovery time will exceed our **15-minute RTO target**, resulting in unrecoverable revenue loss.

2. Manual Failover Bottleneck for Authentication:

User identity and session verification require manual engineer intervention to switch nodes during an incident, introducing operational delay and human-error risk during high-stress outages.

3. Third-Party Payment Network Dependencies:

Outages originating from credit card network connectors (Visa/Mastercard) currently lack automated fallback routing, creating an external dependency that bypasses internal recovery controls.

3. Recommended Action Plan & Resource Request

To ensure business continuity and alignment with our SLA commitments, the GRC team recommends prioritizing the following initiatives for Q3/Q4:

© **Initiative A: Multi-Region Global Database Deployment** *Deploy AWS Aurora Global Database with multi-region automatic storage replication under 1 second.*

○ **Outcome:** Guarantees compliance with <15 min RTO and near-zero RPO targets.

© **Initiative B: Automated Failover Scripting (Infrastructure-as-Code)**

Automate failover triggers for User Authentication services.

○ **Outcome:** Eliminates manual engineering bottlenecks and reduces RTO from 1 hour to <10 minutes.

© **Initiative C: Multi-Acquirer Routing Implementation**

Develop automated fallback logic for card processing networks.

○ **Outcome:** Mitigates third-party downtime exposure by routing transactions around degraded processors.

4. Next Steps

Upon executive approval, engineering and GRC will finalize the technical architecture plan, update the Disaster Recovery (DR) playbook, and conduct a simulated failover exercise in Q4 2026.